# OncoSemiDx

## Transforming Cancer Detection in India

Where Nature Meets Nanotech & AI

Affordable, portable early cancer diagnostics powered by CMOS nanotechnology and AI biomarker analysis

onco-semi-dx.vercel.app

CMOS Chip  AI Analysis  Biosensor

# The Problem

## India's Cancer Crisis

**1.4M**
New cancer cases diagnosed annually in India

**70%**
Detected at advanced stages, limiting treatment options

**₹1 Lakh**
Average cost per diagnostic test, unaffordable for most

**900,000**
Preventable deaths yearly due to late detection

Early detection could save hundreds of thousands of lives every year.
Our mission is to make cancer screening accessible to every Indian,
regardless of location or economic status.

# The Current Barrier

## The Old Way: Centralized Labs

Current high-precision tests (like NGS) are slow, complex, and centralized. They are inaccessible for mass screening.

- ✓ **Cost:** ₹20,000 - ₹1,00,000+
- ✓ **Time:** 1-3 Weeks
- ✓ **Access:** Major Urban Hospitals Only

## The OncoSemiDx Way: Point-of-Care

We bring diagnostics to the patient. Our device is built for clinics, not just specialized labs, enabling proactive screening.

- ✓ **Cost:** Drastically Lower (Target <$50)
- ✓ **Time:** ~30 Minutes
- ✓ **Access:** Any Clinic, Anywhere

# Technology Deep Dive

AI Model Performance & Architecture

### Deep Learning + XGBoost

Ensemble architecture combining neural networks with gradient boosting

### >92% Accuracy

Consistent performance across multiple cancer types and cohorts

### >95% Sensitivity

Minimizing false negatives for critical early detection

### 30 Minutes

Total processing time from sample to result

# How It Works

## End-to-End Workflow

### 1 Data Collection

Biomarker sample (blood/saliva)

### 2 CMOS Biosensor

MOS chip analysis and detection

### 3 AI Processing

Deep learning algorithms

### 4 Graph Analysis

Biosensor pattern interpretation

### 5 Prediction Output

Results in ~30 min with >92% accuracy

Our integrated system combines hardware and AI to deliver accurate results in minutes, not days, making early cancer detection accessible to all Indians.

# Our Solution

Integrated, Affordable, Point-of-Care

## 1. AI-based Gene Analysis

- Advanced algorithms identify cancer biomarkers with >95% accuracy
- Specialized for Gene of Interest (GOI) identification
- Continuous learning with expanding dataset

## 2. CMOS Chip Diagnostics

- Metal-oxide-semiconductor nanotechnology
- Ultra-sensitive detection at nanoscale
- Cost-effective fabrication enabling affordable tests

## 3. Biosensor Graph Analysis

- Real-time waveform interpretation
- Noise filtering and signal enhancement
- Pattern recognition for reliable diagnostics

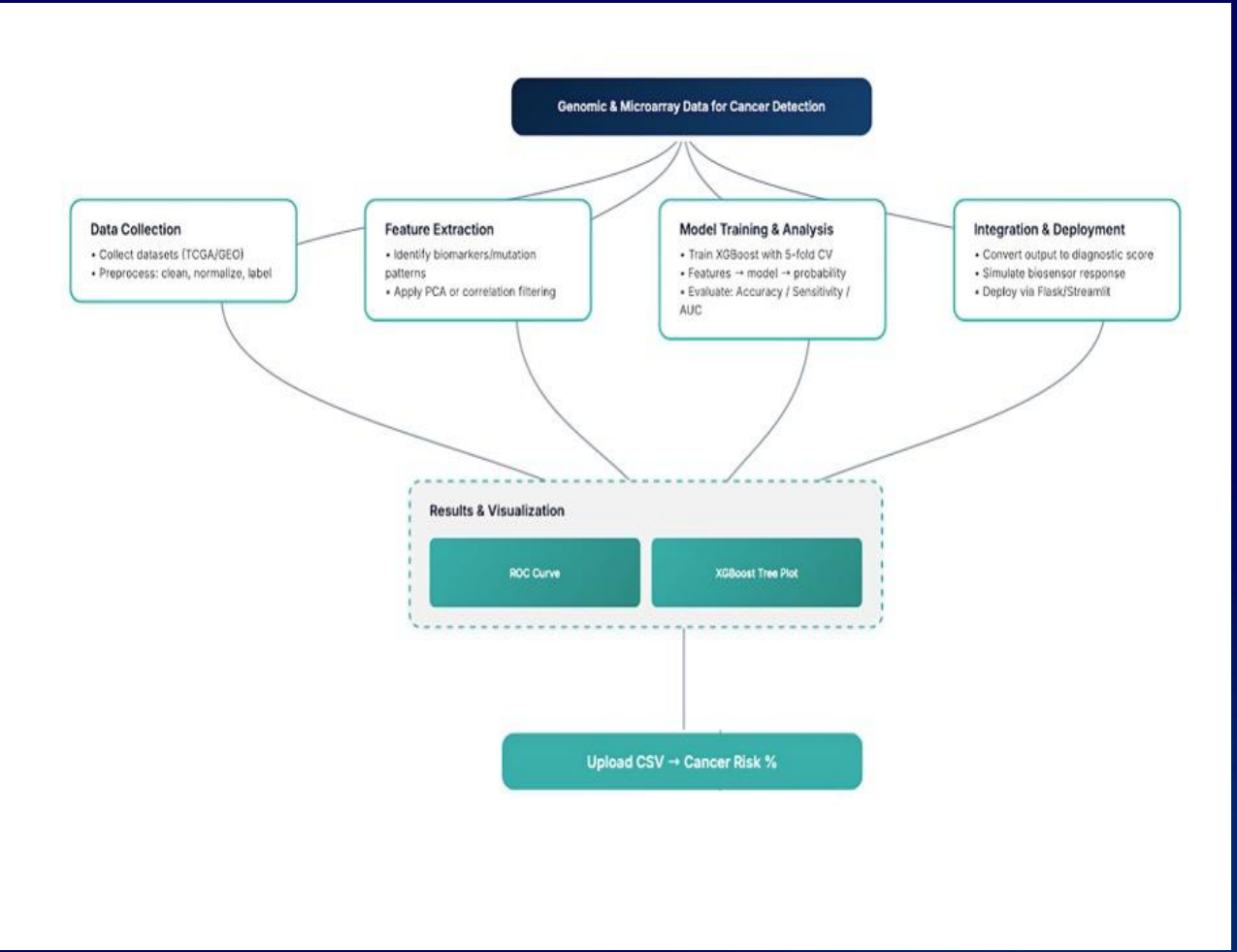
### Portable Point-of-Care Device

Complete diagnostic solution delivering results in approximately 30 minutes, requiring minimal infrastructure and

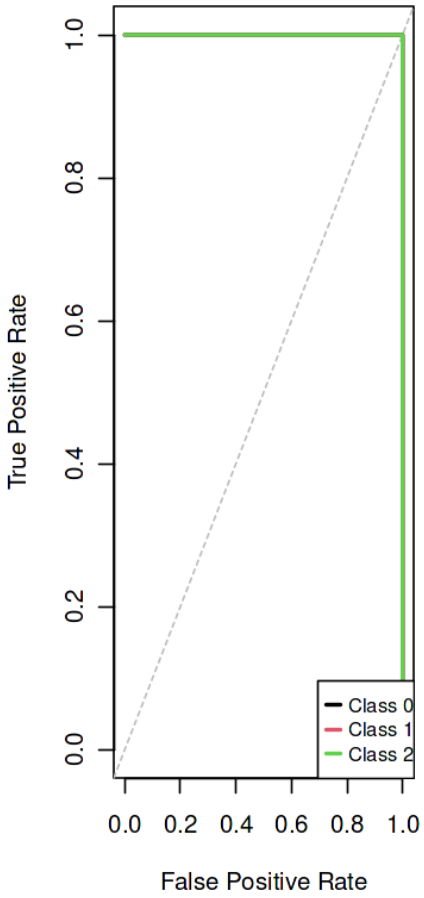
# Tech 1: 92% Accuracy Gene AI

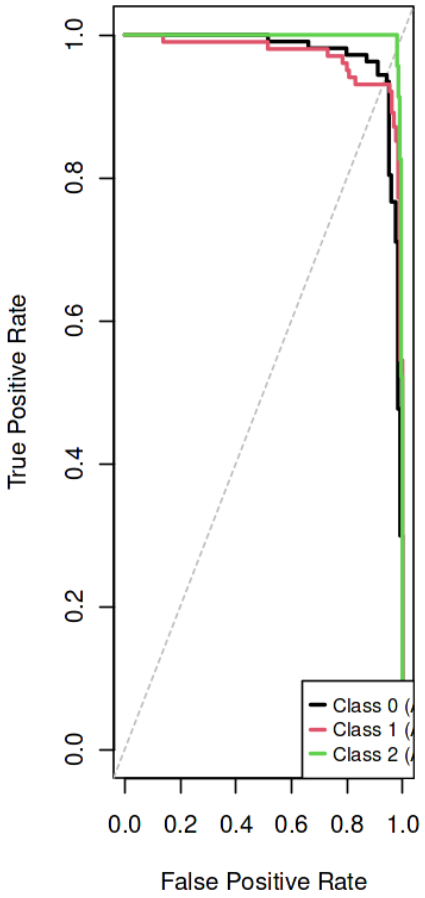
## How It Works

- ✓ We trained a deep learning model (CNN) on vast genomic datasets (e.g., The Cancer Genome Atlas).
- ✓ The AI learns to identify complex, non-obvious patterns in specific "Genes of Interest" (GOI).
- ✓ It can distinguish cancerous signatures from healthy or benign mutations with high fidelity.
- ✓ **Result: 92% accuracy** validated against diverse, independent datasets for early-stage detection.



XGBoost Top 20 Features

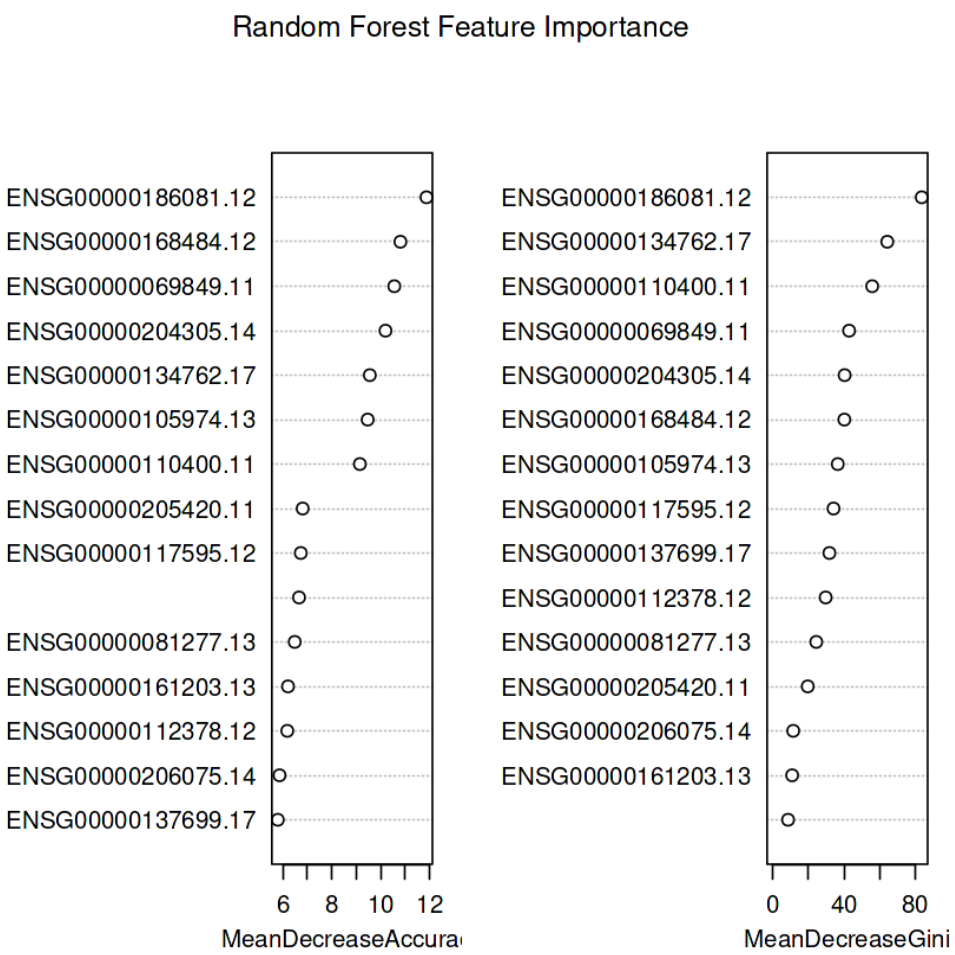


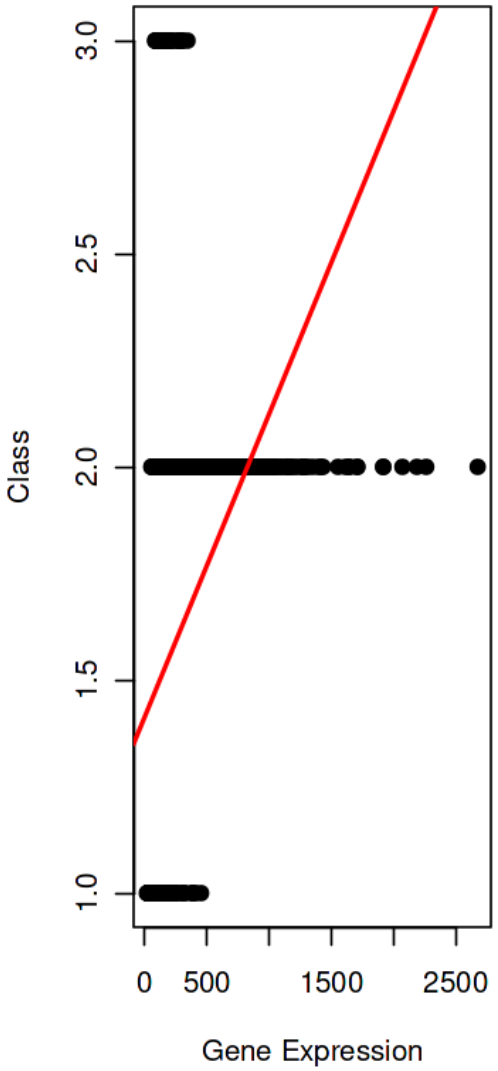

ROC Curves

```
Confusion Matrix:
          Reference
Prediction   0   1   2
         0 102  10   1
         1   4  90   1
         2   1   1  21
```

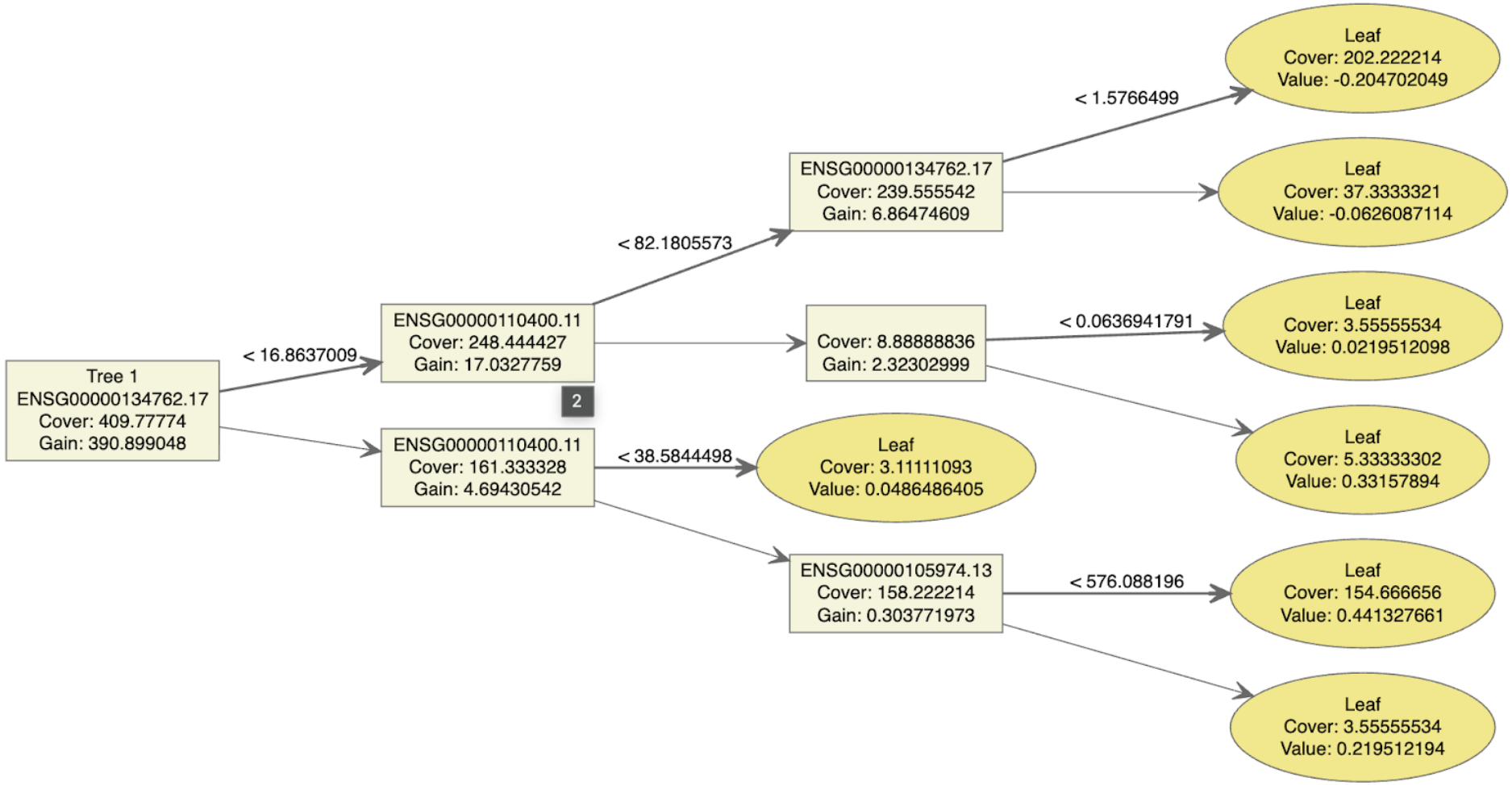
Confusion Matrix



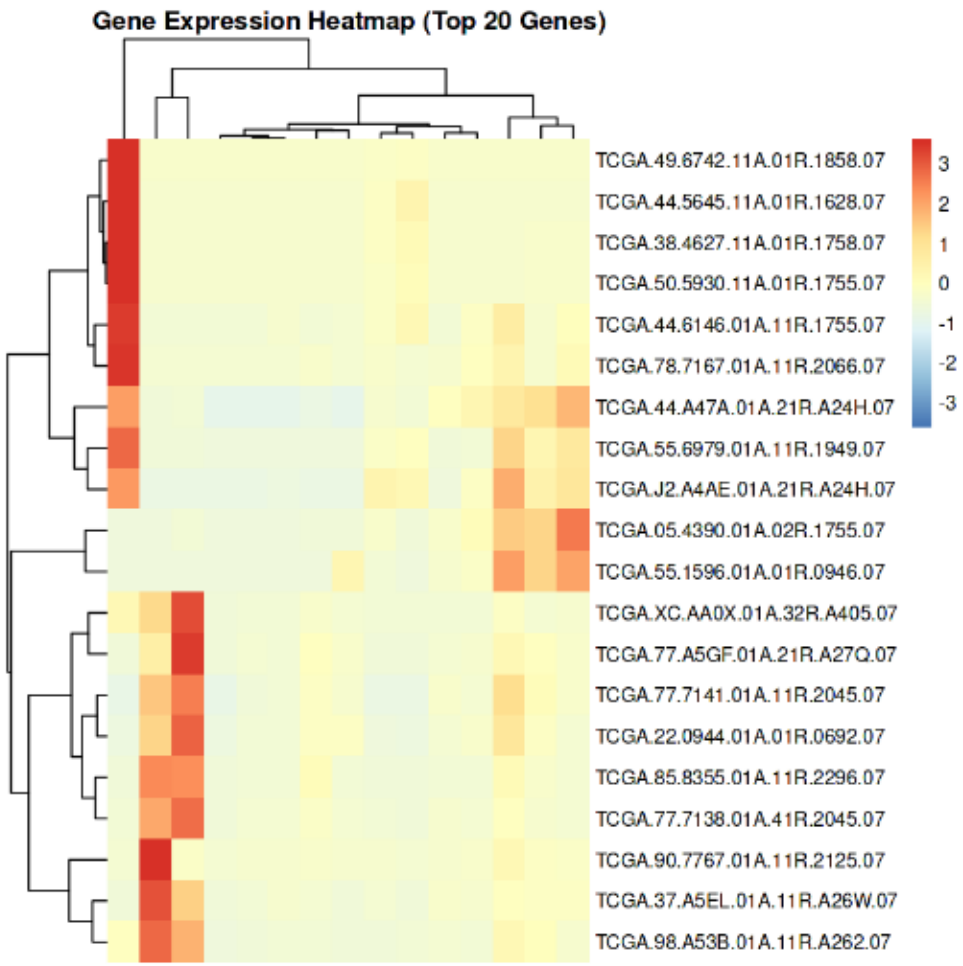

Random Forest

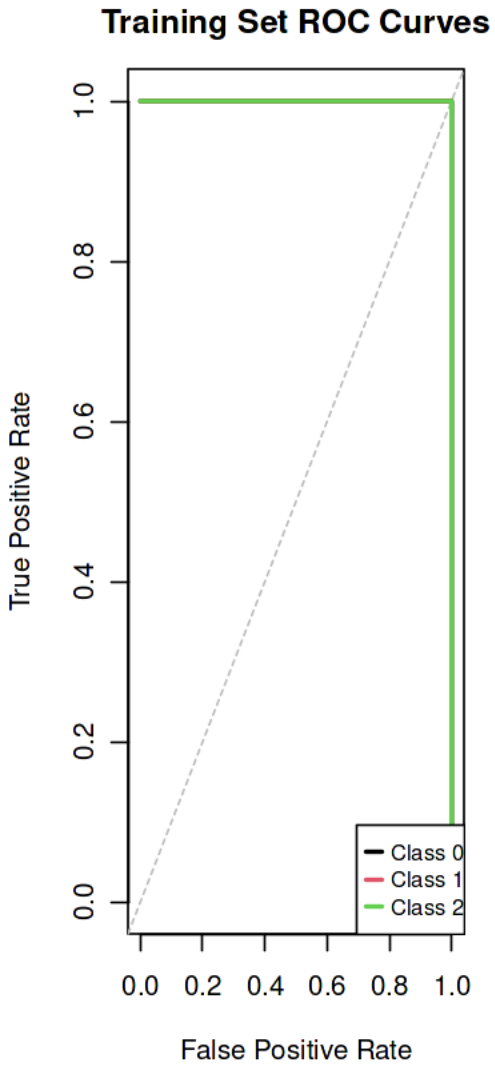


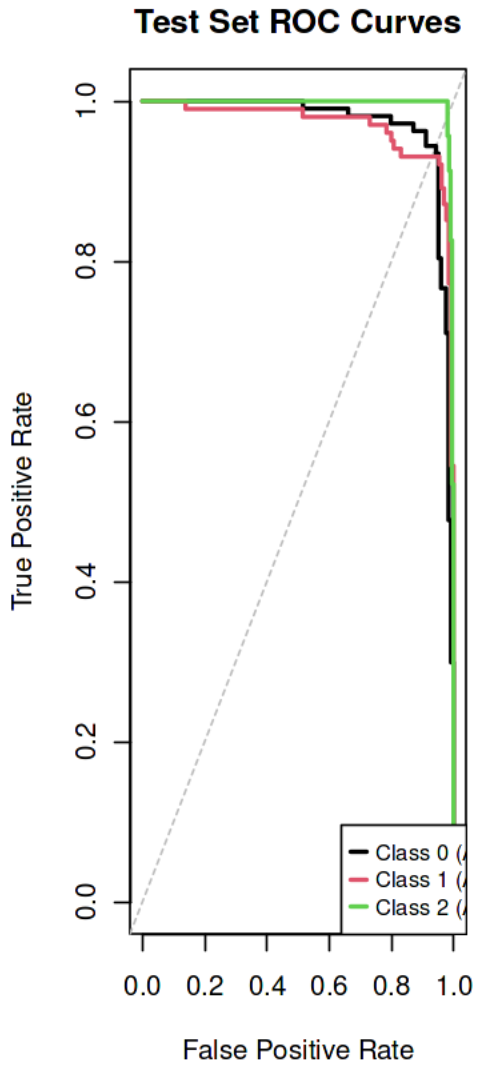

Linear Regression




XGBoost Tree Plot


Gene Expression Heatmap (Top 20 Genes)
TCGA.49.6742.11A.01R.1858.07
TCGA.44.5645.11A.01R.1628.07
TCGA.38.4627.11A.01R.1758.07
TCGA.50.5930.11A.01R.1755.07
TCGA.44.6146.01A.11R.1755.07
TCGA.78.7167.01A.11R.2066.07
TCGA.44.A47A.01A.21R.A24H.07
TCGA.55.6979.01A.11R.1949.07
TCGA.J2.A4AE.01A.21R.A24H.07
TCGA.05.4390.01A.02R.1755.07
TCGA.55.1596.01A.01R.0946.07
TCGA.XC.AA0X.01A.32R.A405.07
TCGA.77.A5GF.01A.21R.A27Q.07
TCGA.77.7141.01A.11R.2045.07
TCGA.22.0944.01A.01R.0692.07
TCGA.85.8355.01A.11R.2296.07
TCGA.77.7138.01A.41R.2045.07
TCGA.90.7767.01A.11R.2125.07
TCGA.37.A5EL.01A.11R.A26W.07
TCGA.98.A53B.01A.11R.A262.07
3
2
1
0
-1
-2
-3


Training Set ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Class 0
Class 1
Class 2


Test Set ROC Curves
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1.0
Class 0
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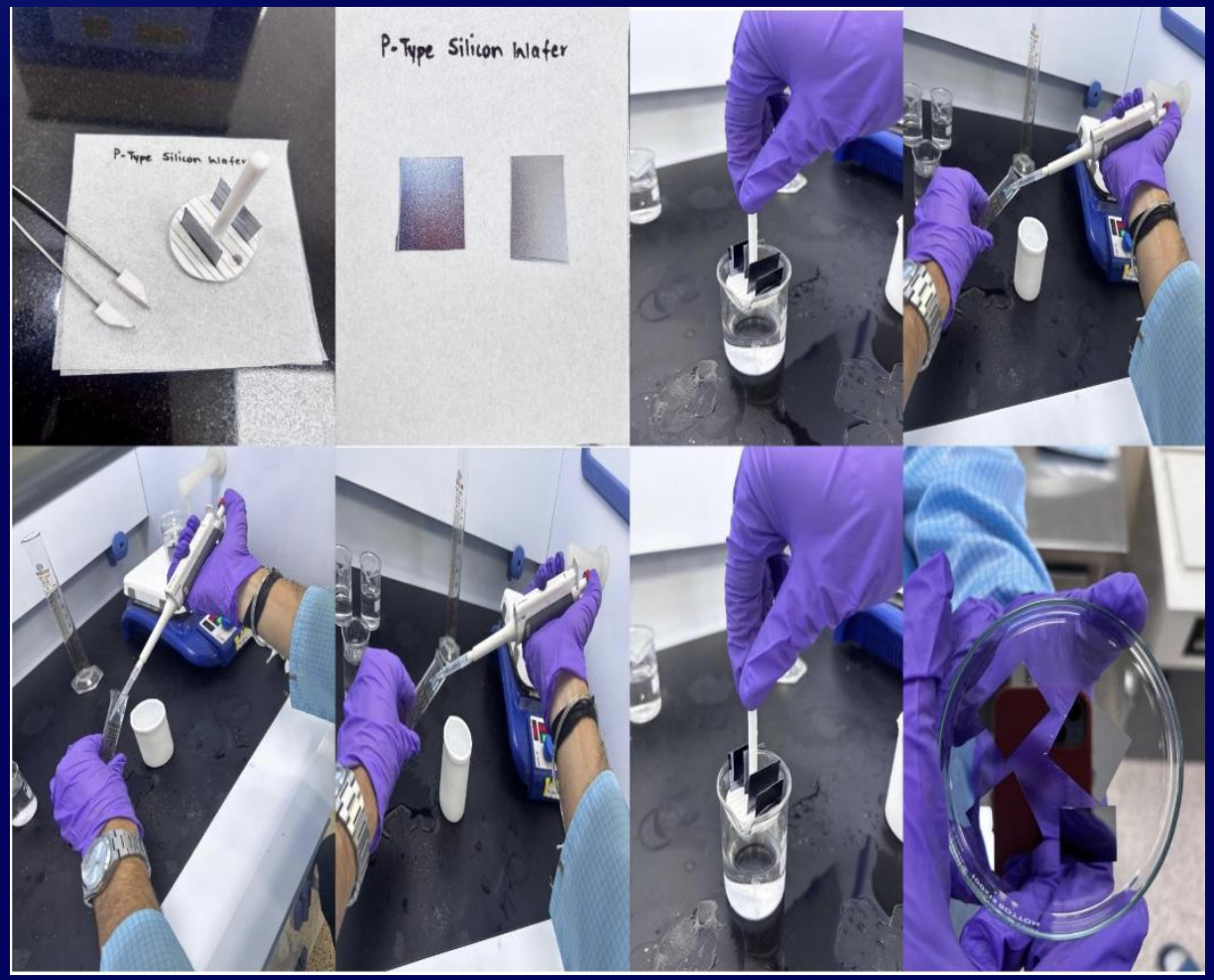
Class 2

# Tech 2: The CMOS "Lab-on-a-Chip"



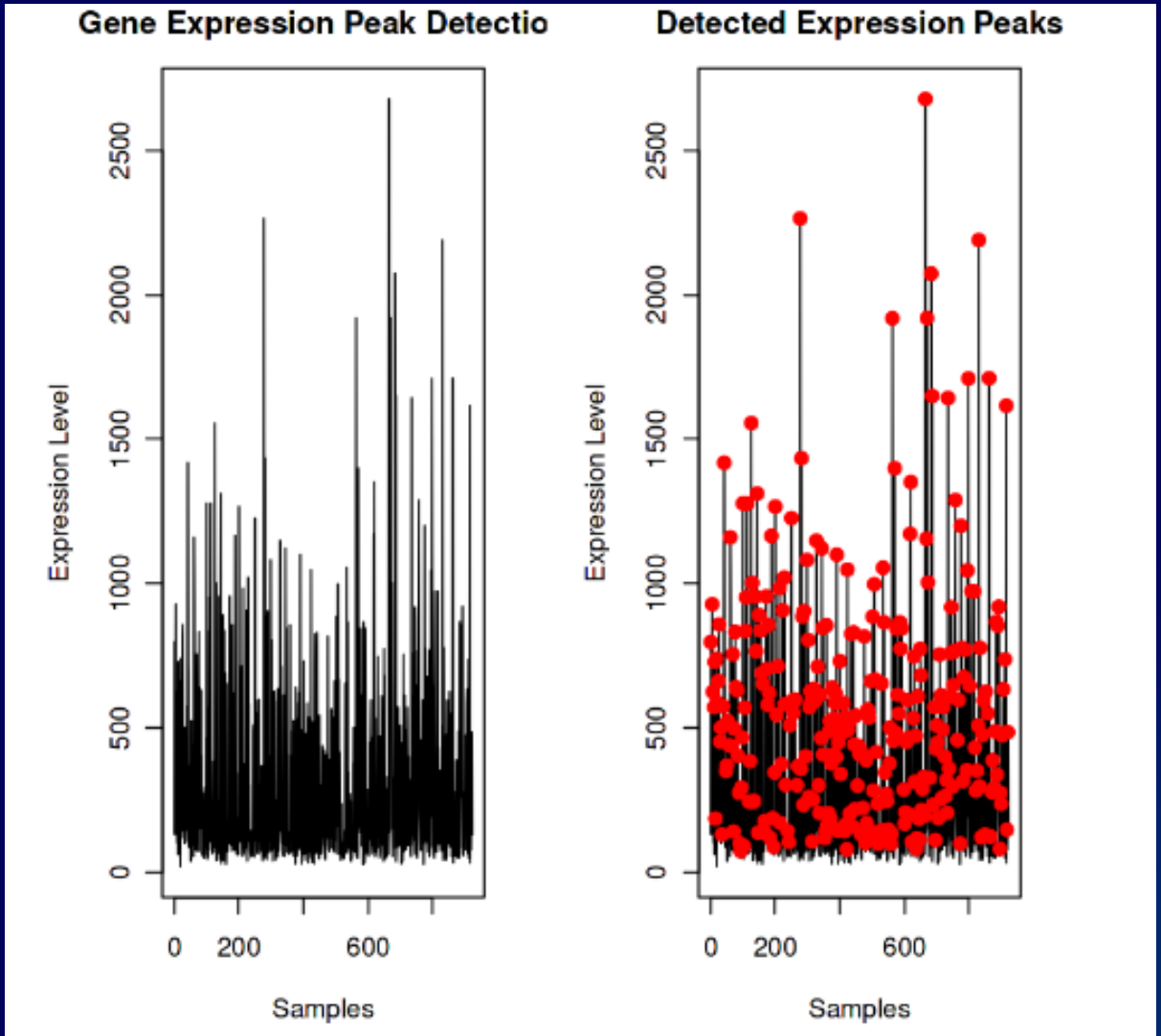

## How It Works

- ✓ The silicon chip surface is "functionalized" with receptors to "catch" specific cancer biomarkers (e.g., target DNA).
- ✓ When a biomarker binds, it changes the electrical properties (capacitance/resistance) of the transistor gate beneath it.
- ✓ This creates a tiny, measurable digital signal.
- ✓ **Why CMOS?** It uses standard, globally-available semiconductor manufacturing. It's scalable, reliable, and incredibly cheap at scale.

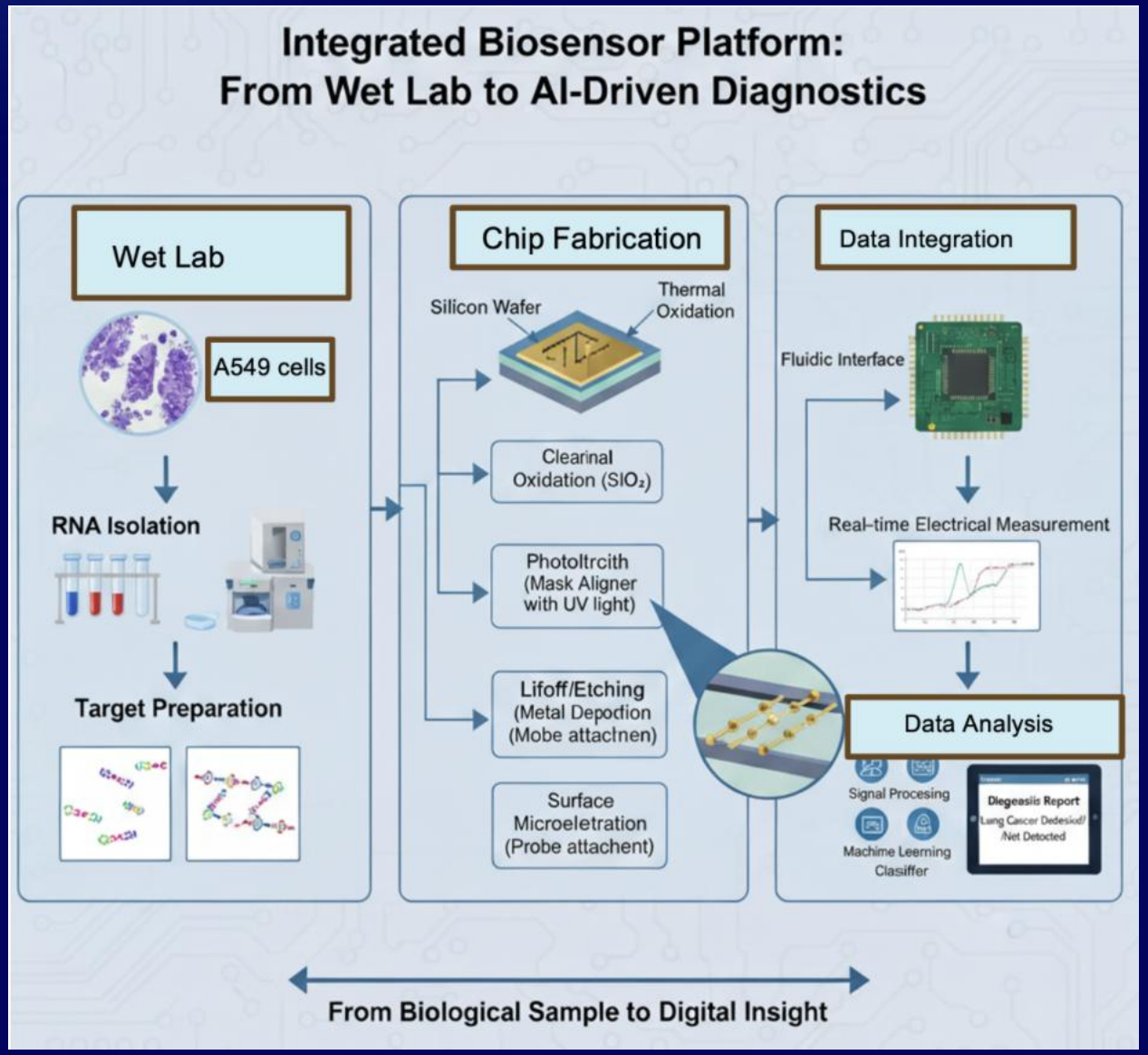
# Tech 3: AI Noise Graph Analysis

## Finding the Signal in the Noise

- ✓ In early-stage cancer, biomarker concentrations are tiny. The "signal" from the chip is often buried in background electronic and thermal "noise."
- ✓ Our second AI is a signal processing network. It's trained to recognize the unique *electrical signature* of a true binding event versus random noise.
- ✓ This allows us to detect cancer far earlier and with much higher sensitivity than traditional methods.



# The Integrated Point-of-Care Device




## Why The Device Matters

The "box" is just as important as the chip. Our device integrates all three technologies into one self-contained, handheld unit:

- ✓ **The Cartridge:** A disposable slot for the CMOS chip.
- ✓ **The Processor:** An on-board chip that runs both the Gene AI and the Noise AI.
- ✓ **The Interface:** A simple screen for operation in any clinic, no specialist required.

# Competitive Advantage

Why We Win

| | OncoSemiDx | Traditional Labs |
|---|---|---|
| Cost | **₹10,000** | ₹1,00,000 |
| Speed | **30 minutes** | 7-14 days |
| Accuracy | **>92%** | Comparable |
| Accessibility | **Portable** | Centralized |
| Infrastructure | **Minimal** | Expensive |

**90% more affordable** with **300x faster results**, while maintaining clinical-grade accuracy and bringing diagnostics directly to underserved communities across India.

# Impact Metrics

## Making Cancer Detection Accessible for All

**10x**

Higher Survival Rates

Through early-stage detection and timely intervention

**90%**

Cost Reduction

Making diagnostic testing affordable for all Indians

**100%**

Rural Accessibility

Portable device deployable in any healthcare setting

OncoSemiDx's technology has the potential to transform cancer diagnostics in India, democratizing access for 1.4 billion people regardless of location or economic status.

# Business Model

## Multiple Revenue Streams

### Device & Test Strips

₹50,000 hardware + ₹1,000/test consumables

High margin recurring revenue model

### Software Licensing

₹10,000/month cloud AI platform

Subscription-based analytics & reporting

### Government Integration

National Health Programs

Large-scale deployment through state tenders

### Scalable Manufacturing

Local production facilities

Cost efficiency and rapid distribution

### Revenue Projections

| Year 1 | Year 3 | CAGR |
|---|---|---|

# How OncoSemiDx Compares

## Cost Per Test

## Time to Result

# Go-To-Market Strategy

## Phased Rollout Plan

### Phase 1
0-6 months

**Pilot Program**

- Deploy in 5-10 Tier-2 city hospitals
- Complete ICMR-aligned validation studies
- Finalize engineering builds & QA processes

### Phase 2
7-12 months

**Initial Scale**

- Scale to 100+ centers across key states
- CDSCO regulatory submission
- Early state tender applications

### Phase 3
Year 2

**National Expansion**

- Scale to 1000+ centers nationwide
- National health program integration
- Establish distributor & service network

### Phase 4
Year 3+

**Rural & International**

- Rural PHC & telemedicine center deployment
- Expand to South Asia & Africa
- International regulatory approvals

**Market Enablement**

KOL engagement · Clinician training · Distribution partnerships · Support infrastructure

# Traction & Milestones

## Our Progress & Roadmap

### Completed

**AI Model Developed** ✅

Achieved >92% accuracy across diverse sample sets

**Prototype Device Completed** ✅

Functional CMOS chip + biosensor integration

**Regulatory Pathway Identified** ✅

CDSCO classification & submission requirements

**Pilot Partnerships** ✅

3 medical centers in discussions for initial pilots

### Upcoming

**CDSCO Approval** 🎯

Target: 12 months

**First 100 Devices Deployed** 🎯

Target: 18 months

**Future Milestones**

- Pan-India coverage in 36 months
- International expansion in 48 months
- Advanced multi-cancer screening in 24 months

**Critical Path to Launch**

- Clinical validation with ICMR
- Manufacturing scale-up
- Quality management system
- Government partnerships

# Our Value Proposition

## Radically Cost-Effective

By leveraging proven, low-cost CMOS technology, we can reduce diagnostic test costs by an order of magnitude. This makes routine screening feasible for the first time.

## Accessible & Rapid

Our portable, point-of-care device eliminates the long wait times and logistical hurdles of centralized labs. It's built for the clinic, not just the specialized hospital.

# Vision — 5-Year Roadmap

## Democratizing Cancer Detection for Every Indian

### Year 1
Foundation

**Launch**

- Complete regulatory approvals
- Deploy first 100 devices
- Establish manufacturing pipelines

### Year 2
Growth

**Scale**

- Deploy in 1,000+ centers
- Screen 1M+ patients
- Expand to 10+ states

### Year 3
Expansion

**Dominance**

- Deploy in 5,000+ centers
- Screen 5M+ Indians annually
- Reduce late detection by 25%

### Year 4
Innovation

**Regional Leader**

- Enter first 3 SE Asian countries
- Deploy in 8,000+ centers
- Screen 8M+ patients annually

### Year 5
Market Leader

**Vision Achieved**

- India's #1 cancer diagnostics
- 10,000+ centers, 5 countries
- 10M+ screenings, 50% fewer late detections

### Long-term Vision

Democratize early cancer detection for 1.4 billion Indians and beyond, transforming how healthcare is delivered in resource-constrained settings globally through our integrated AI-driven semiconductor platform.

# Our Funding Ask

| Category | Immediate Needs (Current Month/Initial Run) (₹) | 18-Month Scaling & Deployment (₹) | Total Allocation (₹) |
|---|---|---|---|
| 1. Gene AI Model | ₹40,000 | ₹1,00,000 | ₹1,40,000 |
| 2. Semiconductor Chip | ₹50,000 | ₹1,50,000 | ₹2,00,000 |
| 3. Graph Analysis | ₹20,000 | ₹50,000 | ₹70,000 |
| 4. Other Mis. R&D / Regulatory | ₹40,000 | ₹50,000 | ₹90,000 |
| Total | ₹1,50,000 | ₹3,50,000 | ₹5,00,000 |

## Use of Funds

This capital will be used to achieve our critical next milestones:

- Finalizing device-level clinical trials and validation.
- Securing semiconductor fabrication (fab) partners.
- Scaling initial production and funding our go-to-market team.

# Thank You

OncoSemiDx: Making early cancer detection a reality for everyone.

ADHAAR KOHLI

ARUSHI JAIN

SHRESTH MISHRA